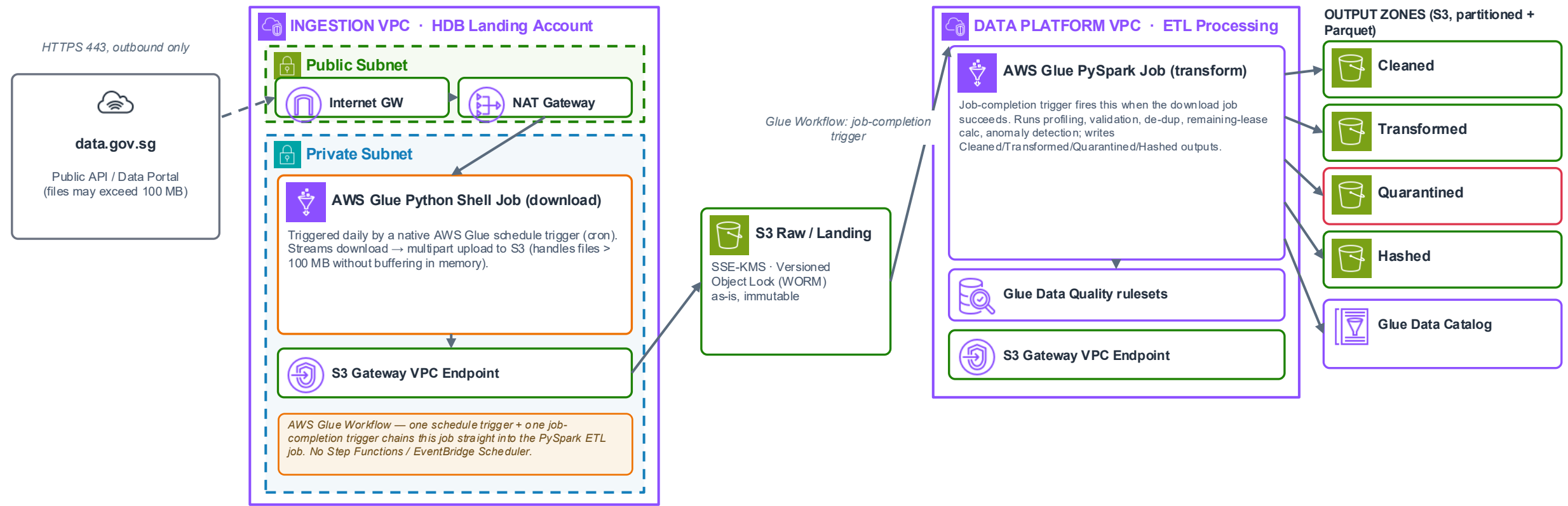


HDB Resale Flat Data Platform

Part 2 — Architecting Data Ingestion & Data Exploitation Solution Patterns (AWS)

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Diagrams built with the official AWS Architecture Icons

Data Ingestion Architecture



CROSS-CUTTING SECURITY, IDENTITY & MONITORING · applies across Ingestion VPC, Data Platform VPC, S3 zones, and Output zones

IAM Role

Secrets Manager

KMS CMK

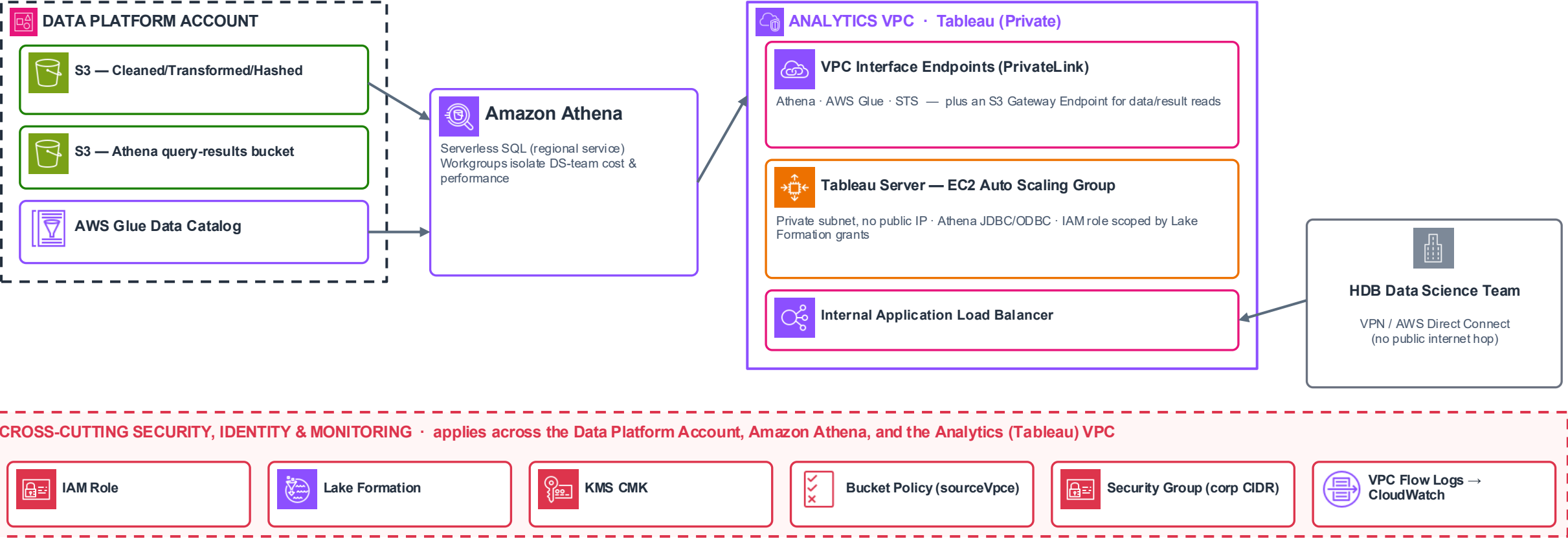
CloudTrail

CloudWatch

SNS

- Glue Python Shell job (private subnet) fetches data.gov.sg over HTTPS via the public subnet's NAT Gateway/IGW — outbound only.
- The file streams straight into S3 Raw/Landing (via an S3 Gateway Endpoint) as an immutable, versioned, as-is copy.
- A Glue Workflow job-completion trigger starts the PySpark job — validates, dedupes, computes lease, flags anomalies, hashes records.
- Outputs land in four S3 zones (Cleaned/Transformed/Quarantined/Hashed) and are catalogued in Glue Data Catalog for Athena.
- IAM, Secrets Manager, KMS, CloudTrail, CloudWatch and SNS apply as shared controls across the whole pipeline, not per box.

Data Exploitation Architecture



NODE BORDER KEY (icons are the official AWS Architecture Icons)

- Compute
- Storage
- Analytics
- Integration / Monitoring
- Security / IAM
- Private Subnet

- Amazon Athena (serverless, regional) queries the Cleaned/Transformed/Hashed S3 zones directly, using Glue Data Catalog for schema and Lake Formation for fine-grained grants.
- Tableau Server (EC2 Auto Scaling Group, Analytics VPC) reaches Athena, Glue and STS purely through VPC Interface Endpoints (PrivateLink) — never over the public internet.
- HDB Data Science Team analysts reach Tableau via VPN/Direct Connect and an internal ALB; Tableau itself has no public IP.
- IAM, Lake Formation, KMS, bucket policies, Security Groups and VPC Flow Logs apply as shared controls across the Data Platform Account, Athena, and the Analytics VPC.

Design Considerations

Security

- Public data.gov.sg egress isolated to a small ingestion subnet; NAT/IGW never touch the processing or analytics VPCs
- No component that reads/writes data holds a public IP; all AWS-service traffic (S3, Athena, Glue, STS) stays on PrivateLink / Gateway Endpoints
- S3 bucket policies scoped to VPC endpoints (aws:sourceVpce); SSE-KMS on all zones; Lake Formation for column/row-level access
- Least-privilege IAM roles per component; Secrets Manager for credentials; CloudTrail + VPC Flow Logs for full audit trail
- IAM, KMS, CloudTrail and other identity/monitoring controls are shown once as a shared cross-cutting layer in both diagrams, since they apply uniformly across VPCs rather than being duplicated per box

Scalability

- Glue jobs and Athena are serverless — scale automatically with data volume and concurrent Tableau queries
- Tableau Server on an Auto Scaling Group behind an internal ALB absorbs analyst concurrency spikes
- S3 scales storage independently of compute; Athena Workgroups give the DS team isolated concurrency/cost limits
- AWS Glue Workflow triggers (schedule + job-completion) decouple the download and transform jobs without a separate orchestration service to scale

Maintainability

- Entire stack defined as Infrastructure-as-Code (CloudFormation/CDK/Terraform), version-controlled and peer-reviewed
- CI/CD pipeline (CodePipeline/CodeBuild) deploys and tests Glue job scripts and Glue Workflow/trigger definitions
- Glue Data Catalog is the single source of schema truth; Glue Data Quality rules encode Part 1 validation rules as reusable, testable artefacts
- CloudWatch dashboards + SNS alerts give operators one place to observe pipeline health

Performance

- Cleaned/Transformed/Hashed zones stored as partitioned, compressed Parquet (by year-month, town) for Athena partition pruning
- Glue Job Bookmarks enable incremental processing so re-runs only touch new files
- VPC Endpoints avoid NAT Gateway bottleneck/cost for S3, Glue and Athena traffic from the Analytics VPC
- Athena Workgroups + per-query data-scan limits protect shared performance and cost

Assumptions & Notes

Account structure

The Ingestion VPC and Data Platform VPC sit in the same AWS account: AWS Glue Workflows (used for orchestration below) are account-scoped, so both Glue jobs need to be in one account for the schedule/job-completion triggers to chain across them. Network segmentation between them is still enforced at the VPC/subnet level. The Analytics (Tableau) account remains separate, reached only via PrivateLink/regional APIs — never a VPC peer.

Batch cadence

Ingestion runs daily via a native AWS Glue schedule trigger (cron) rather than a separate EventBridge Scheduler, since a single daily batch job doesn't need its own scheduling service.

Orchestration

A single AWS Glue Workflow chains the two ETL steps: a schedule trigger (daily) starts the Python Shell download job, and a job-completion trigger starts the PySpark transform job the moment it succeeds — no Step Functions or EventBridge needed for this linear, two-step pipeline.

>100 MB file handling

Downloads are streamed and multipart-uploaded to S3 from an AWS Glue Python Shell job rather than a plain Lambda, since Lambda's /tmp and payload limits make it a poor fit for large, memory-heavy downloads.

Athena network path

Athena, Glue Data Catalog and S3 are regional AWS-managed services, not deployed inside a VPC. Privacy of the Tableau↔Athena path is achieved with VPC Interface Endpoints (PrivateLink) in the Analytics VPC plus S3-bucket policies scoped to those endpoints — a Transit Gateway/VPC-peer is only introduced if Tableau needs direct network access to other Data Platform VPC resources.

Access to Tableau

Data Science Team analysts reach Tableau over the corporate network via VPN/Direct Connect and an internal ALB; Tableau is never exposed on a public subnet or public IP.

Data formats

Raw files are retained as-is (CSV, per data.gov.sg) in the Raw zone; Cleaned/Transformed/Hashed zones are converted to partitioned Parquet to optimise Athena scan cost and performance.

Row/column-level security

AWS Lake Formation grants are assumed sufficient to restrict sensitive columns (if any are added later) without needing separate Athena views per role.

Out of scope

Actual coding/IaC for Part 2 was not required by the brief; this deck documents the target-state design only.